Research questions and study design

Preparation for 26 October 2026

Use an LLM to explore research questions and study design using a paper you know well. **A first attempt and questions are enough**; we will continue working in pairs during the session.

What to do

Choose a published paper you know well. Select its research question and short, relevant extracts from the Methods and Results; you do not need to provide the whole paper. Use public material you are permitted to process, leaving out confidential or unpublished research.

Before using an LLM, write two or three sentences about your own view of the design and what you expect the model to suggest or criticise. Then choose **one** of these tasks:

1. **Design a follow-up study.** Identify a question the original paper leaves open. Ask for a study blueprint and assess whether its key design choices fit that question.
2. **Try another design.** Ask how a different design could answer the same question. Explain the gains and losses, and whether the question or comparison changes.
3. **Critique and improve the chosen design.** Ask for specific methodological strengths, weaknesses and improvements. Check each concern against the supplied extracts.

Try and compare

Use a model of your choice and ask it to flag missing information. Check its suggestions against the paper and your own reasoning. Compare the response with your initial expectation: what would you accept, adapt or reject, and why? Refine your prompt or discuss a doubtful suggestion with the model.

If possible, try the same task with LUMC Copilot and a local model. Give both the same short extracts and prompt. Compare usefulness, errors, speed and practical limitations.

Optional extension

If a public protocol is available, provide the relevant protocol sections first and save the model's initial assessment. Then add the article extracts. Ask for agreements, differences and uncertainties, with exact passages from both documents. Check the findings yourself. A difference is not automatically an inappropriate protocol deviation: consider the protocol version, amendments and any explanation. If these are unavailable, leave the conclusion open.

What to bring

Bring your paper reference, selected extracts, initial expectation, prompts and output. Keep a brief log of the model and version, relevant settings and any changes to your approach. Bring a few notes on what worked or failed, plus one design decision or unresolved question to discuss. Partial results are welcome.